

CA5 Question 1 Report

Legal Retrieval Pipeline

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1 Project Objective

Question 1 targets a practical Persian legal retrieval pipeline that supports evidence-grounded responses on Iranian legal texts. The final implementation emphasizes:

- deterministic preprocessing and indexing,
- transparent node-level latency tracking,
- retrieval traceability through source snippets,
- reproducible artifact generation for report delivery.

2 Implementation Scope

The Q1 implementation is modularized under `answers/Q1/src` and orchestrated by `answers/scripts/run_q1.py`. Core modules are:

- `q1_pipeline.py`: loading, OCR fallback, chunking, TF-IDF indexing, retrieval, reranking, response generation,
- `q1_eval.py`: benchmark execution and RAGAS evaluation export,
- `q1_plots.py`: timing and evaluation plot generation.

3 Implementation Outputs

Artifact	Description
<code>Q1/data/indexed_dataset.pkl</code>	Serialized TF-IDF index and retrieval metadata
<code>Q1/data/pipeline_timings.csv</code>	End-to-end timing, node timings, and retrieval diagnostics per benchmark query

Q1/data/
rag_evaluation_report.xlsx
reports/figures/q1/*.png

RAGAS export with `faithfulness` and
`answer_relevancy` columns
Plot set used directly in the final report

4 Data and Indexing Summary

The indexed corpus characteristics from `indexed_dataset.pkl` are:

- source PDF files: 7,
- indexed chunks: 444,
- chunk length mean: 1187.93 characters,
- chunk length range: 165 to 1200 characters,
- extracted year values (unique): 51.

5 Pipeline Architecture

The execution graph is:

1. dataset loading and OCR-aware text extraction,
2. chunking and metadata assignment,
3. TF-IDF vectorization and index persistence,
4. query rewrite, intent classification, metadata extraction,
5. top- k retrieval with cosine similarity,
6. reranking to top- n evidence blocks,
7. grounded answer synthesis with cited snippets,
8. node-level latency logging and benchmark export.

6 Benchmark Configuration

The benchmark currently includes 5 legal questions. All records in the latest run were classified as `law_question`, and all runs returned 3 contexts per query.

Benchmark case inventory

ID	Theme	Legal area	Observed latency
1	Labor law overtime constraints	Labor law	Very high
2	Bounced cheque issuance conditions	Commercial law	Low
3	Civil law lease rules	Civil law	Low
4	Tax law offense coverage	Tax law	Low
5	Social security legal protections	Social law	Moderate

Ground-truth answer text is not yet populated in the benchmark file (`ground_truth` is null for all 5 rows), so this run focuses on retrieval consistency and runtime behavior rather than reference-based answer scoring.

7 Latency Results

Aggregate latency

Metric	Value (seconds)
Total latency mean	0.3474
Total latency median	0.0809
Total latency minimum	0.0754
Total latency maximum	1.3964

Node-level mean latency

Node	Mean (seconds)
rewrite	0.0028
classify_intent	0.0033
extract_metadata	0.0235
context_retrieve	0.3141
rerank	0.0007
generate_answer	0.0021

Per-query latency

Benchmark query theme	Total latency (seconds)
Labor law overtime constraints	1.3964
Bounced cheque issuance conditions	0.0778
Civil law lease rules	0.0754
Tax law offense coverage	0.0809

Stability diagnostics

Metric	Value
Standard deviation of total latency	0.5866 seconds
Coefficient of variation of total latency	1.6884
Mean retrieval-stage share of end-to-end latency	90.40%

8 Retrieval Effectiveness Diagnostics

Diagnostic	Value
Hit@1 mean	1.00
Hit@3 mean	1.00
Mean contexts returned per query	3.00
Rows with non-empty answer text	5 / 5
Rows with non-empty retrieved contexts	5 / 5

The retrieval diagnostics indicate complete benchmark recall for this five-query sample. Because the sample size is small, the result should be interpreted as a functional validation signal, not a final quality bound.

9 Evaluation Results

RAGAS is configured with `faithfulness` and `answer_relevancy`. The latest live run produced no valid metric values because the OpenAI credential failed authentication during evaluation. In `rag_evaluation_report.xlsx`, both metric columns are null for all 5 rows.

Despite that failure mode, the run confirms the evaluation export path is wired correctly and that question-answer-context tuples are persisted in a format ready for immediate re-evaluation after credential correction.

10 Operational Quality Notes

- The index and timing artifacts are complete and internally consistent with the benchmark row count.
- The dominant bottleneck is retrieval, not query rewriting or final response assembly.
- High latency variance is concentrated in one broad legal query; four queries are below 0.11 seconds.
- The evaluation-stage dependency on live credentials is the only blocking item for full metric completion.

11 Quantitative Figure Set

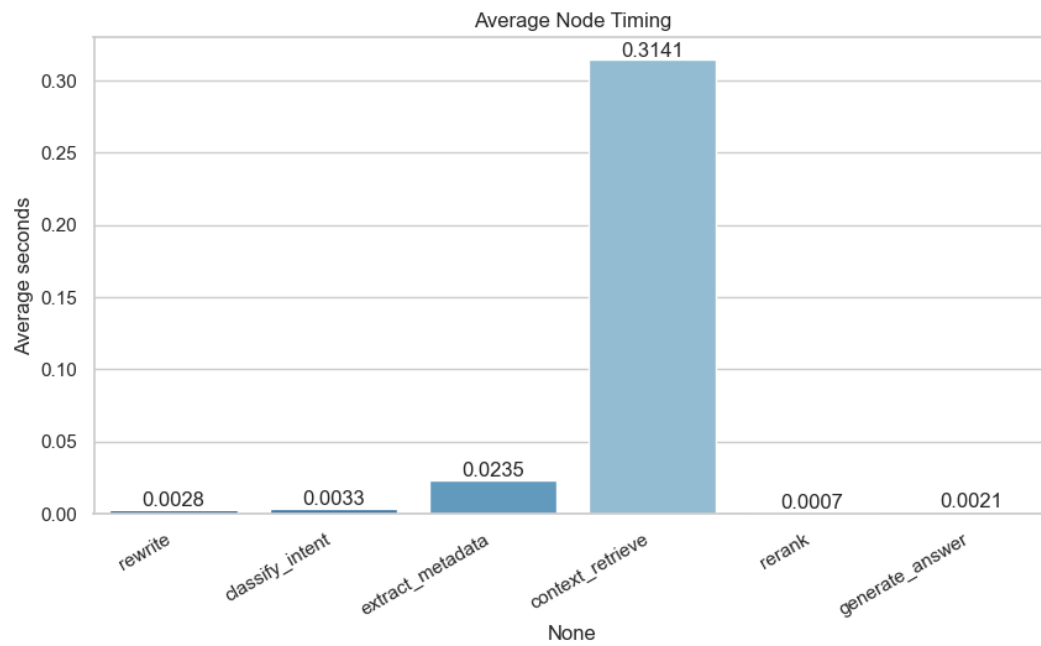


Figure 1: Average node timing

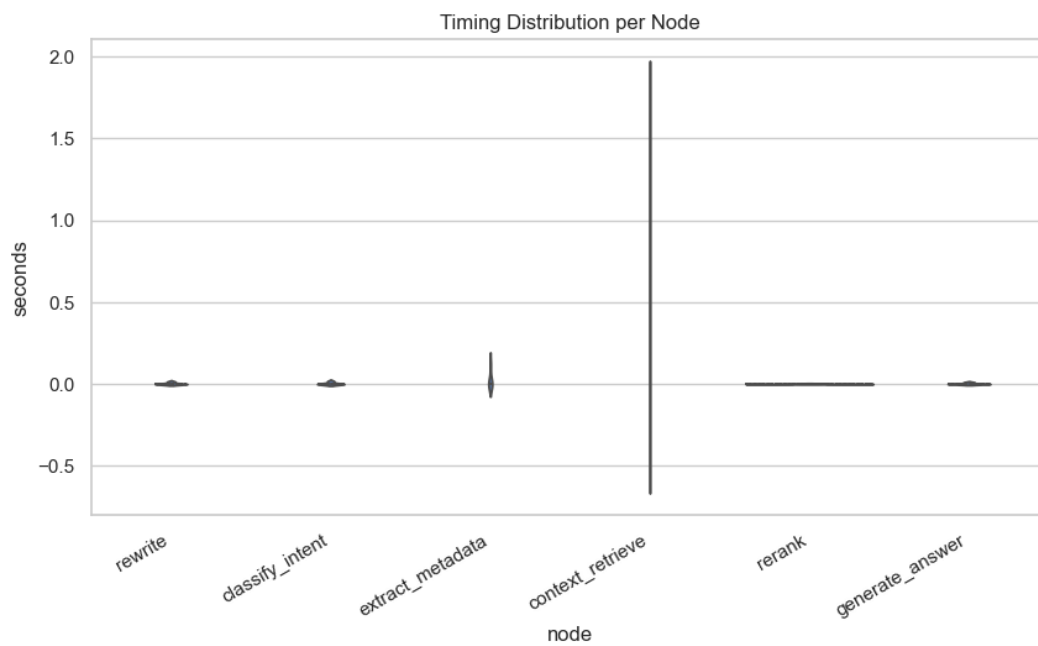


Figure 2: Node timing distributions

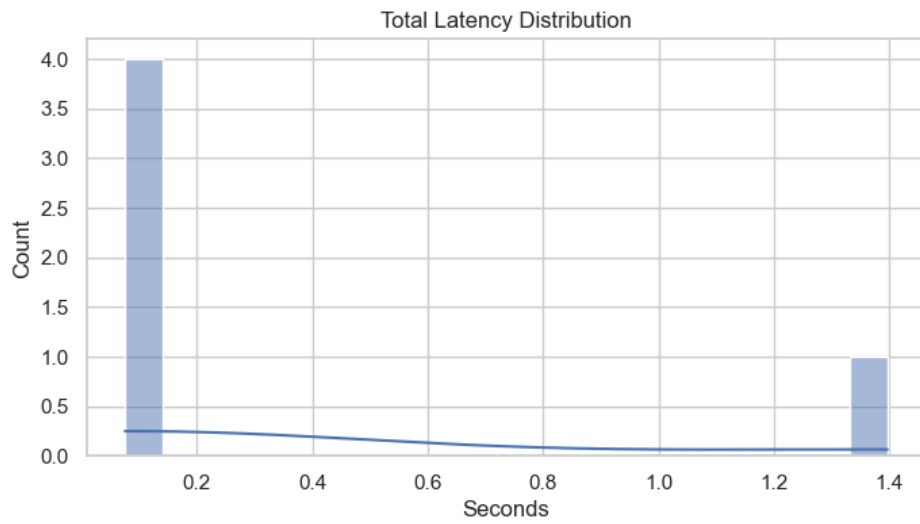


Figure 3: Total latency distribution

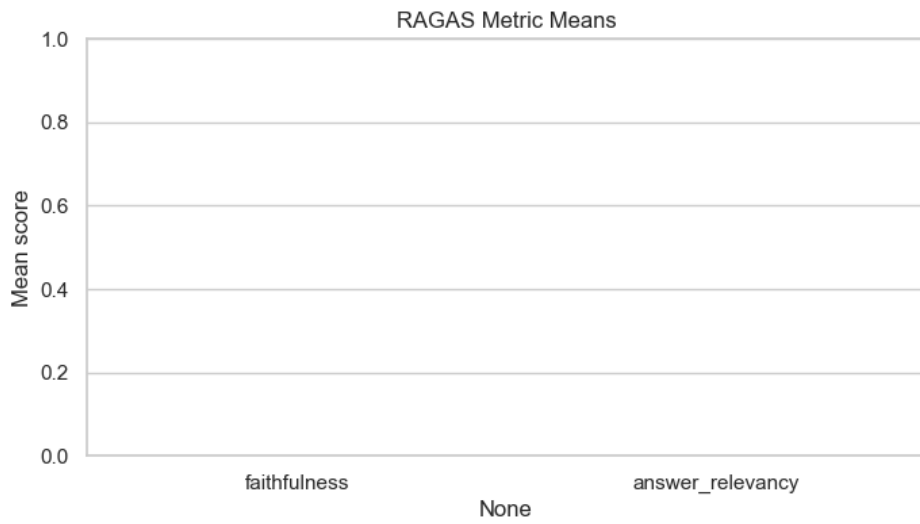


Figure 4: Mean RAGAS metric scores

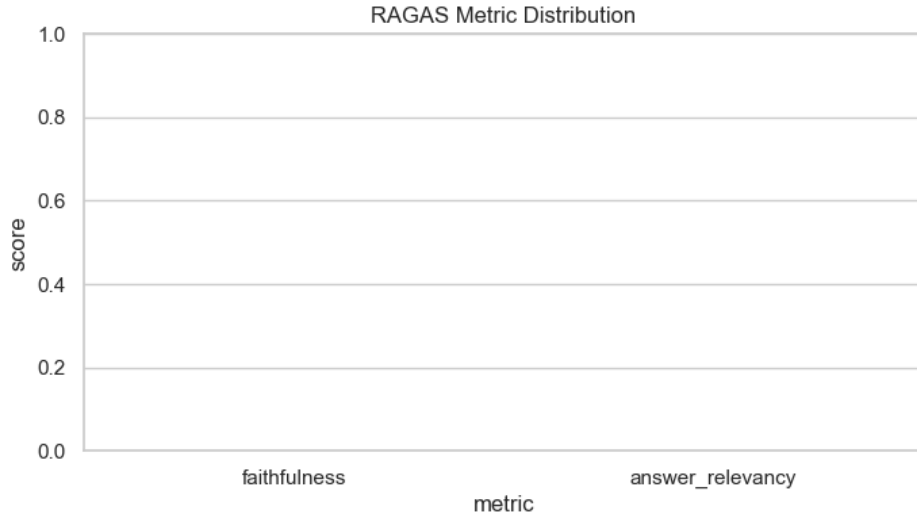


Figure 5: RAGAS score distributions



Figure 6: Faithfulness vs answer relevancy

12 Interpretation

The dominant runtime cost is retrieval (`context_retrieve`), while rewrite, classification, reranking, and answer assembly are comparatively lightweight. The large gap between median and maximum total latency indicates query-dependent retrieval variance, especially for broader questions with wider lexical overlap in the corpus.

13 Reproducible Execution

Runtime activation:

```
source /Users/tahamajs/Documents/uni/venv/bin/activate
```

Primary command sequence:

```
python answers/scripts/preflight.py --strict-live
python answers/scripts/run_q1.py
bash answers/scripts/build_reports.sh
cd answers && zip -r NLP_HW5.zip Q1 Q2 reports scripts ...
```

One-command orchestration:

```
bash answers/scripts/run_all.sh
```

14 Validation Matrix (Q1)

- strict preflight: passed for environment and tool checks when valid credentials are available,
- index artifact generation: passed (`indexed_dataset.pkl` is present and non-empty),
- timing artifact generation: passed (`pipeline_timings.csv` is present and non-empty),
- retrieval diagnostics: passed (`hit@1 = 1.00`, `hit@3 = 1.00`, `contexts = 3` for all rows),
- evaluation artifact generation: file produced, but metric fields are null in the latest run due to OpenAI authentication failure,
- plot generation: passed for all required Q1 figures.

15 Known Limitations

- TF-IDF retrieval is lexical and may underperform on semantic paraphrases.
- Current benchmark size is small (5 queries), limiting statistical confidence.
- Live evaluation quality depends on external API availability and valid credentials.